Course Title	Electro Mechanical Systems	**Course Code**	EE2015
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	**Credit Hrs.**	3+1
Knowledge Area	Power Engineering (KA04)	**Grading Scheme**	Relative
HEC Knowledge Area	Electrical Engineering Core (Breadth)	**Applicable From**	Fall 2025
SDG	9 | Industry, Innovation and Infrastructure	**PBL**	1
Pre-requisite(s)	NL1002, NS1002, EL2004, EE2004	**CEP**	1

Course Objective	To analyze the operation of major types of AC and DC machines, including transformers, generators and motors. Quantitative analysis is carried out for efficient engineering application of electrical machines.

No.	Assigned Program Learning Outcome (PLO)
1	Apply knowledge of mathematics, natural science, engineering fundamentals and Engineering specialization to the solution of complex engineering problems.
2	Identify, formulate, conduct research literature, and analyze complex Engineering problems reaching substantiated conclusions using first principles of mathematics, natural science and engineering sciences.

A = Assignment, Q = Quiz, M = Midterm, F=Final

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Apply the fundamental concepts of magnetism and electromagnetism to understand working of stationary and rotating electromechanical systems	A1, Q1, M1	C3	1
2	Analyze the performance and efficiency of practical single-phase and three-phase transformers using test data and equivalent circuit modeling	A2, M1, F	C4	2
3	Apply operating principles to model and solve performance-related problems of DC machines	Q2, M2, F	C3	1
4	Analyze the construction, operation, and performance characteristics of synchronous machines under various loading and excitation conditions	Q3, F	C4	2
5	Analyze the construction, operation and performance characteristics of induction machines under various starting conditions	A3, F	C4	2

Text Books	**Title**	Electrical Machinery fundamentals
	Author	Stephen J. Chapman
	Publisher	McGraw Hill, 2005, 4th Edition, ISBN 0-07-246523-9
Reference Books	**Title**	Electric Machinery
	Author	Fitzgerald, A.E., Charles Kingsley, Jr. And Stephen Umans
	Publisher	Boston: McGraw Hill, 2002, 6th Edition

Week	Course Contents/Topics	Chapter*	CLO*
01	Fundamentals of Electricity and Magnetism: B-H curve, Hysteresis	1	1
02	Ideal Transformers: Applied and induced voltage, Circuit symbol, problems	2	1
03	Practical Transformers: Equivalent circuit, Polarity test, Taps, Losses, problems	2	1
04	Practical Transformers: Transformer tests, efficiency and voltage regulations, problems	2	1
05	Three phase transformers: Basic properties of 3-phase transformers, ΔΔ, ΔY, YΔ and YY connections, autotransformers	2	1
06	Direct Current Generators: Theory and Equivalent circuits	9	2
07	Direct Current Motors: Principle, Back EMF, construction, equivalent circuits.	9	2
8	Direct Current Motors: Types, Speed control, Torque Speed Curve, Problems	9	2
9	Synchronous Generators: Introduction, Commercial synchronous generators, Number of poles, stator, rotor, Field exciters, Brushless excitation,	4	3
10	Synchronous Generators: Armature Reaction, Equivalent circuit, Efficiency and Voltage Regulation	5	3
11	Synchronous Motors: Introduction, Construction, Starting, Pull-in torque, etc.	6	4
12	Synchronous Motors: Power and torque, Excitation and reactive power,	6	4
13	Three-phase Induction Motors: Introduction, components, Principle of operation, The rotating field, Direction of rotation,	7	5
14	Three-phase Induction Motors: Starting, Torque Speed Curve, Creating High starting torque	7	5
15	Single-phase Induction Motors: Introduction, Construction, Synchronous speed, Torque-speed characteristic, Principle of operation, Capacitor-start motor	11	5
16	Special purpose motors	9	5

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterm (I+II)	30.0%
Final Exam	50.0%